FragPipe-Analyst report

12 August, 2025

Method details

The raw DDA LFQ data files were analyzed using FragPipe to obtain protein identifications and their respective label-free quantification values using using IonQuant. Uploaded combined_protein.tsv file was first filtered for contaminant proteins and then converted to log2 scale. Then, the data was converted to log2 scale, samples were grouped by conditions and missing values were imputed using man method. Protein-wise linear models combined with empirical Bayes statistics were used for the differential expression analyses. The *limma* package from R Bioconductor was used to generate a list of differentially expressed proteins for each pair-wise comparison. A cutoff of the *adjusted p-value* of 0.05 (`params$fdr_correction` method) along with a |log2 fold change| of 1 has been applied to determine differentially expressed proteins in each pairwise comparison.

Quick summary of parameters used:

- Normalization = vsn
- Imputation = man
- Tested comparisons = NPKL_SRRM2_Ab_FANCI_vs_NUCL_NPM1_AB_FANCI
- Adjusted *p-value* cutoff <= 0.05
- Log2 fold change cutoff >= 1

Results

FragPipe result output contains proteins groups of which *2075* proteins were reproducibly quantified.

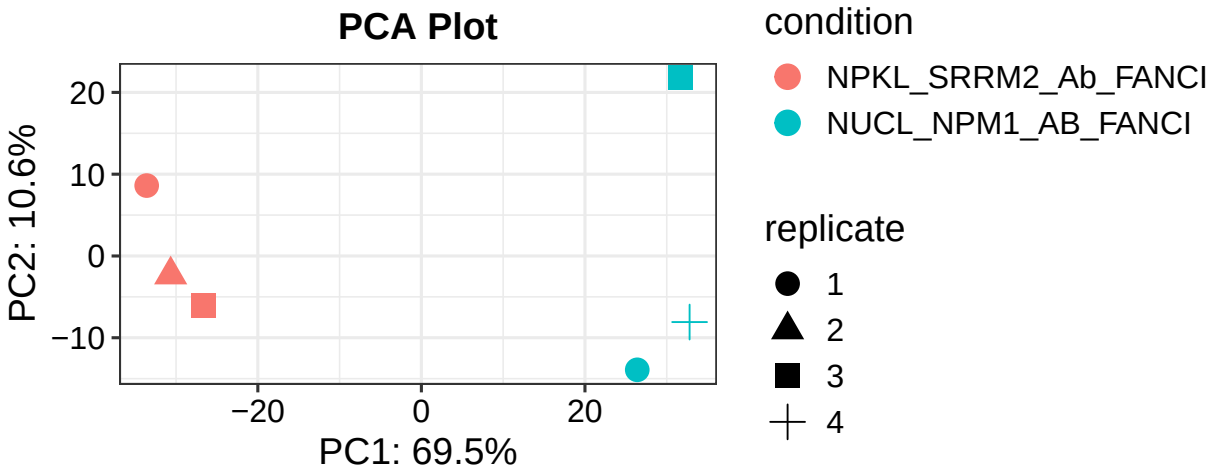
568 proteins differ significantly between samples.

Exploratory Analysis (QC Plots)

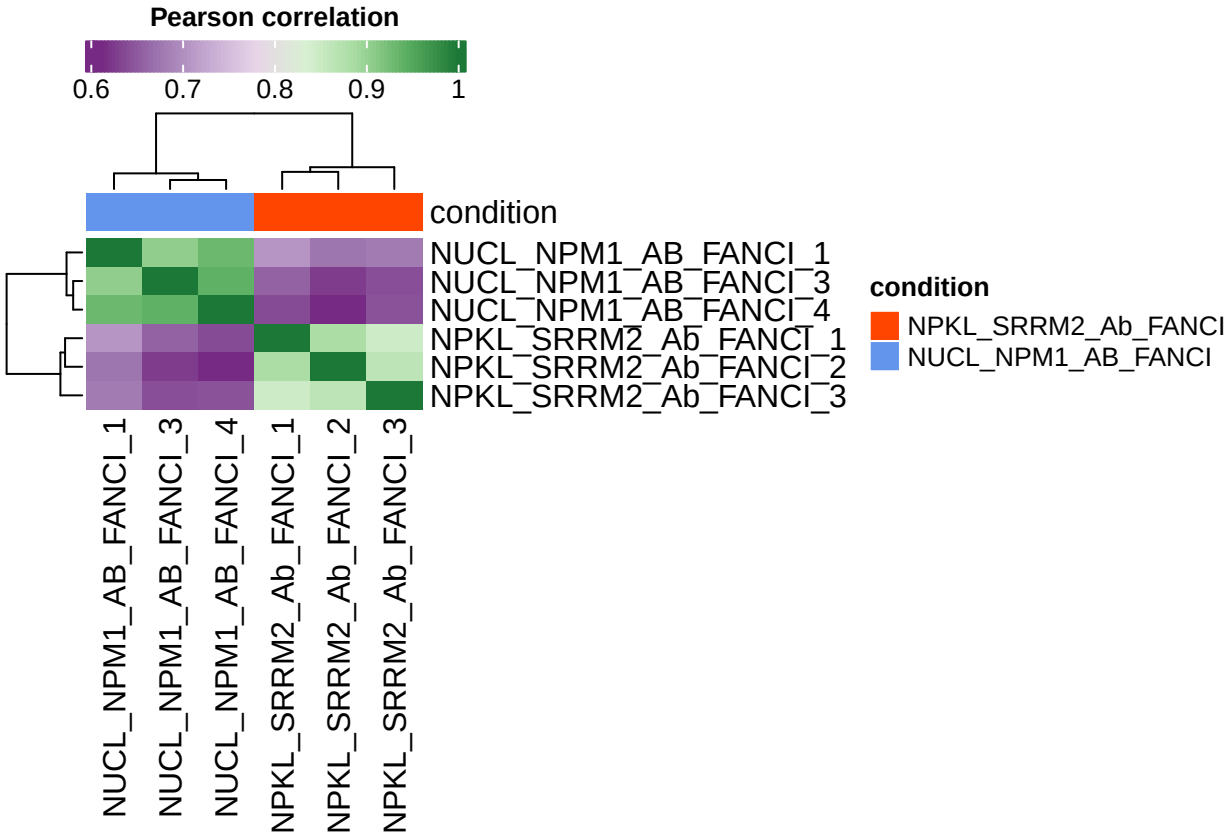
Principle Component Analysis (PCA) plot

```
## Warning: Use of `pca_df[[indicate[1]]]` is discouraged.
## i Use `.data[[indicate[1]]]` instead.
```

```
## Warning: Use of `pca_df[[indicate[2]]]` is discouraged.
## i Use `.data[[indicate[2]]]` instead.
```

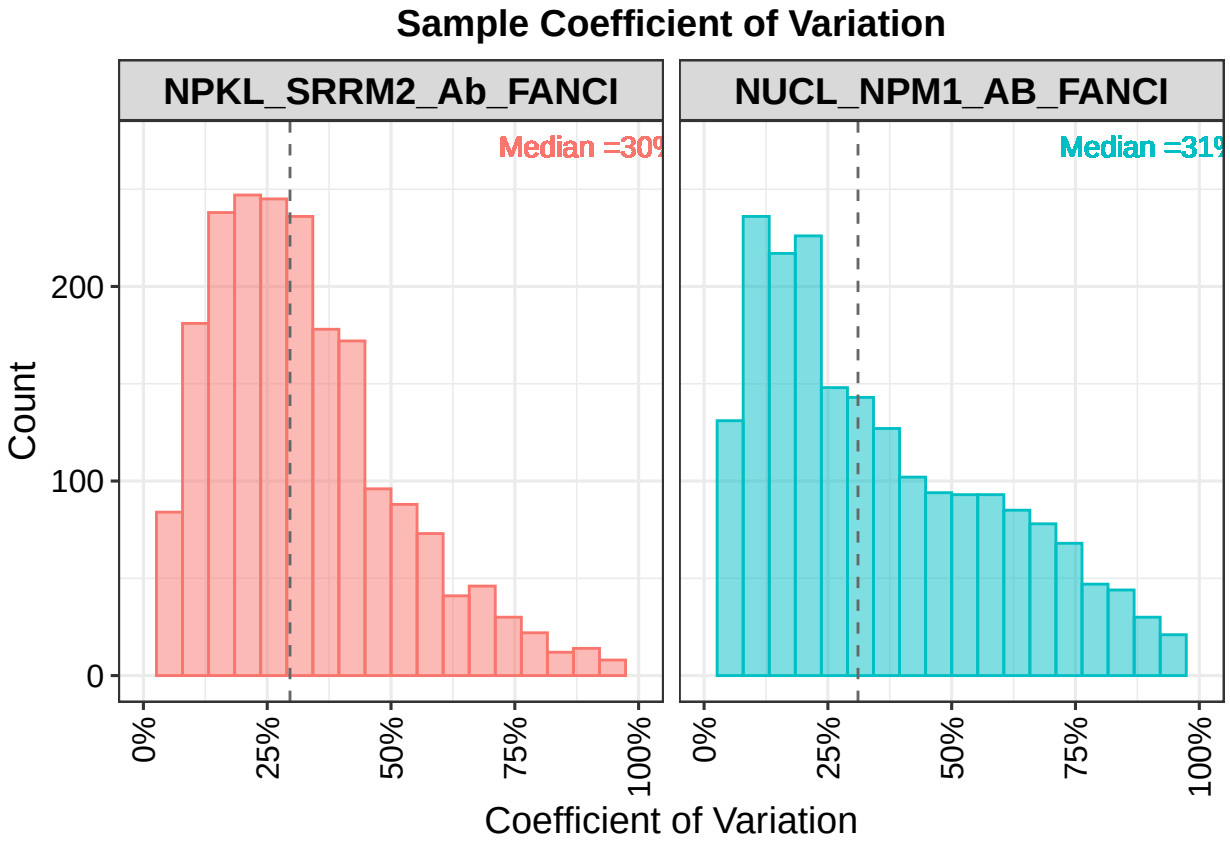


Sample Correlation matrix



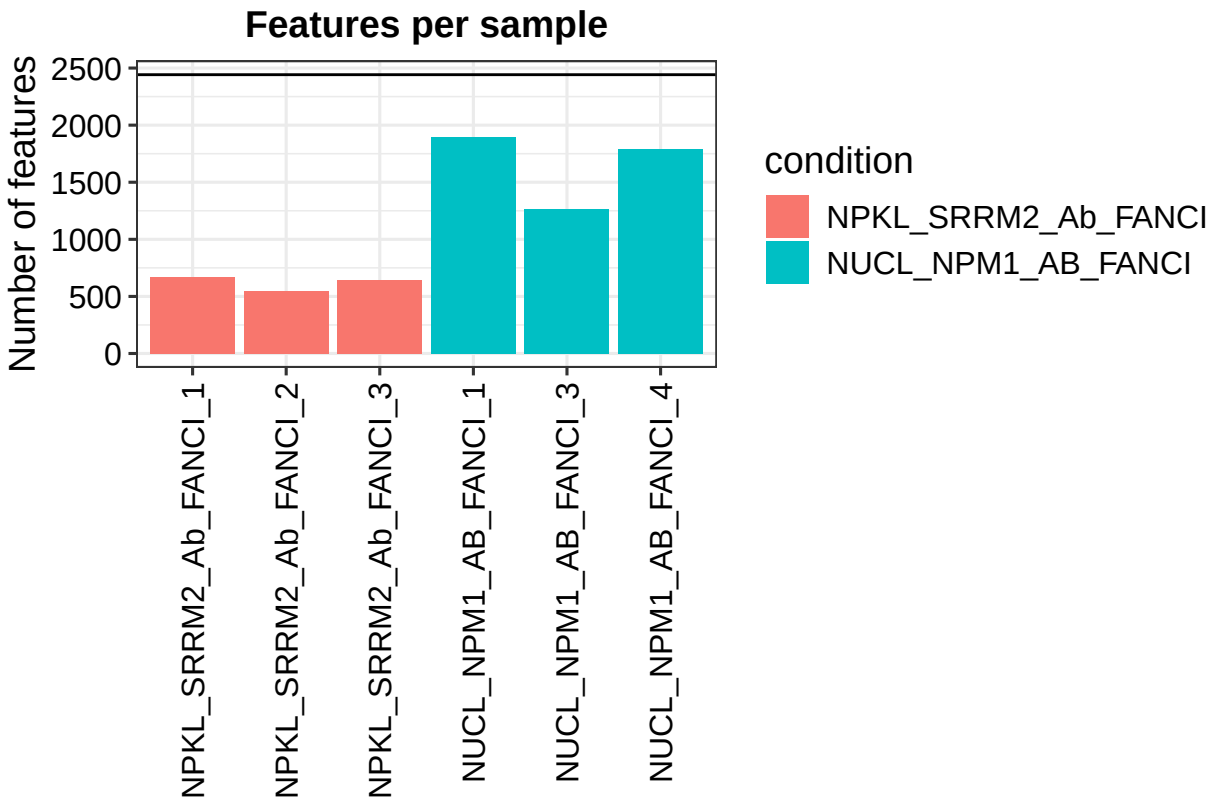
Sample Coefficient of variation (CVs)

```
## `summarise()` has grouped output by 'rowname'. You can override using the `.groups` argument.
```

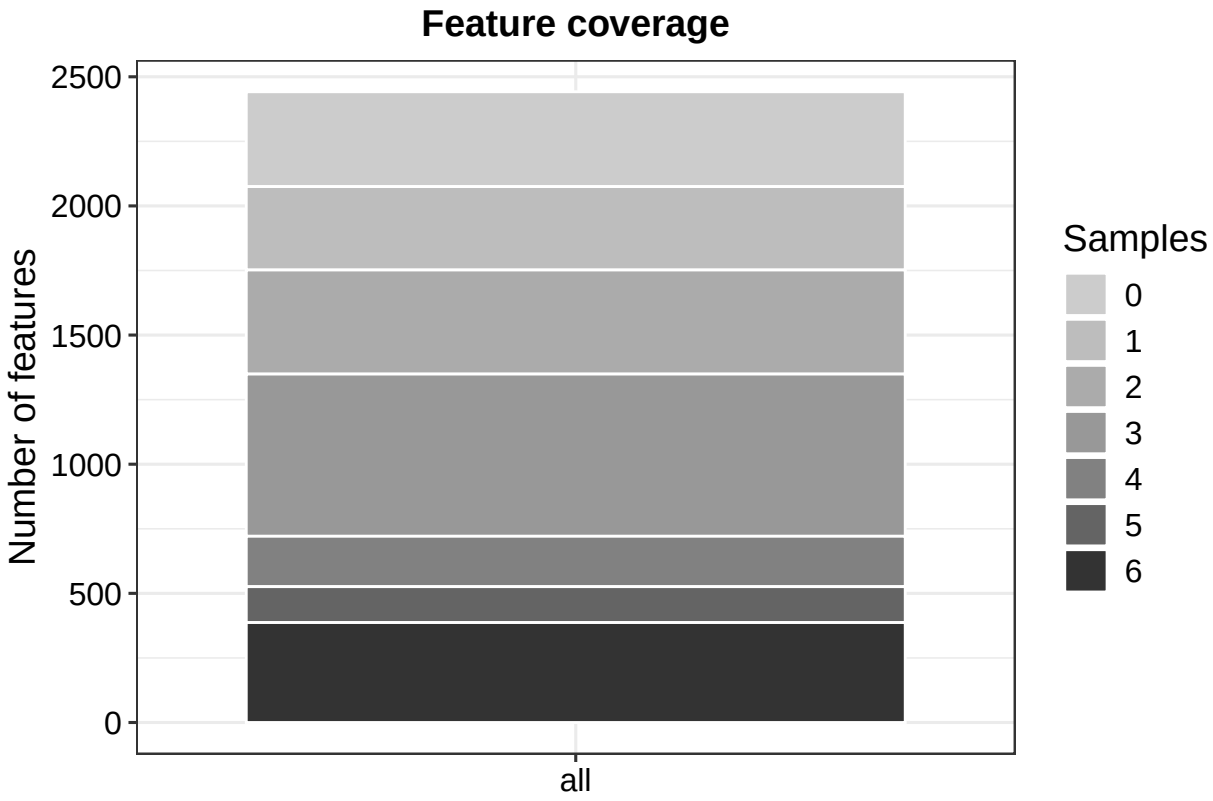


Proteomics Experiment Summary

Protein quantified per sample (after pre-processing).



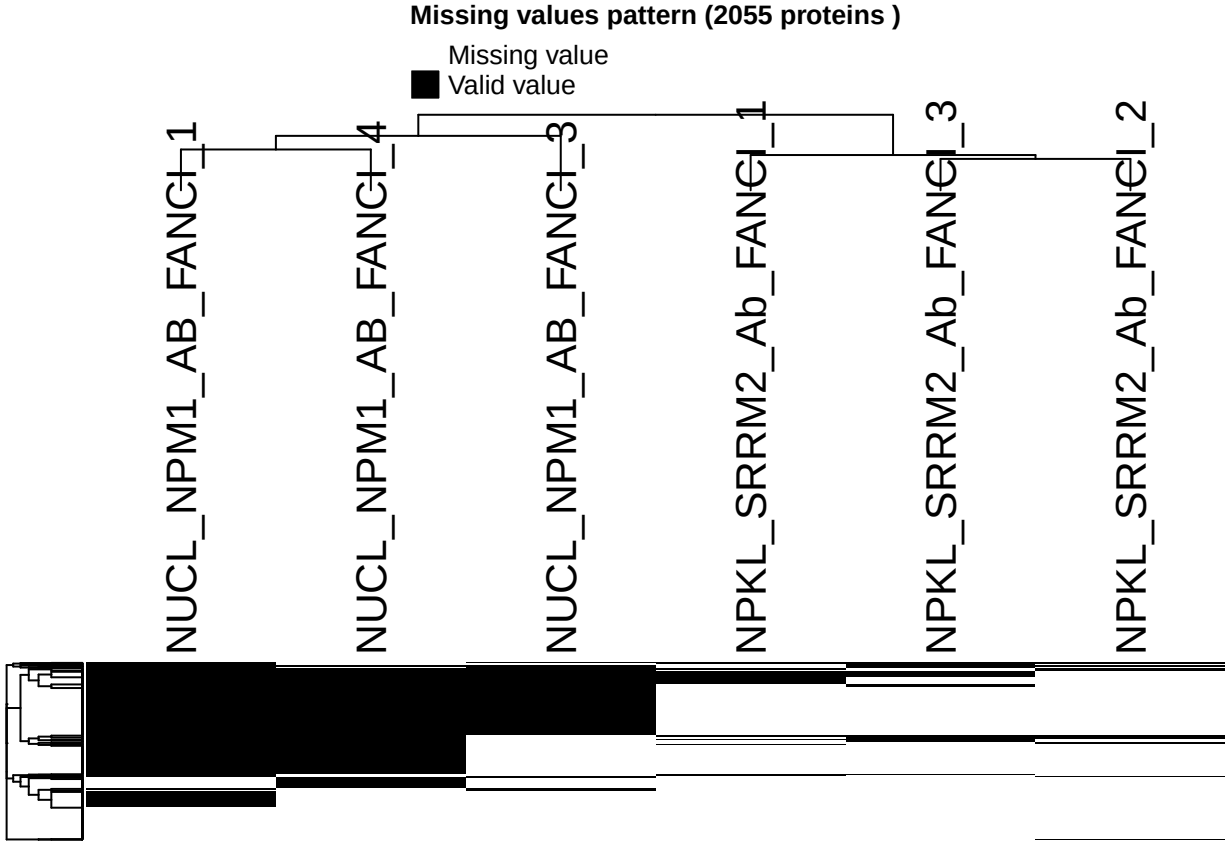
Protein overlap in all samples.



Missing Value handling

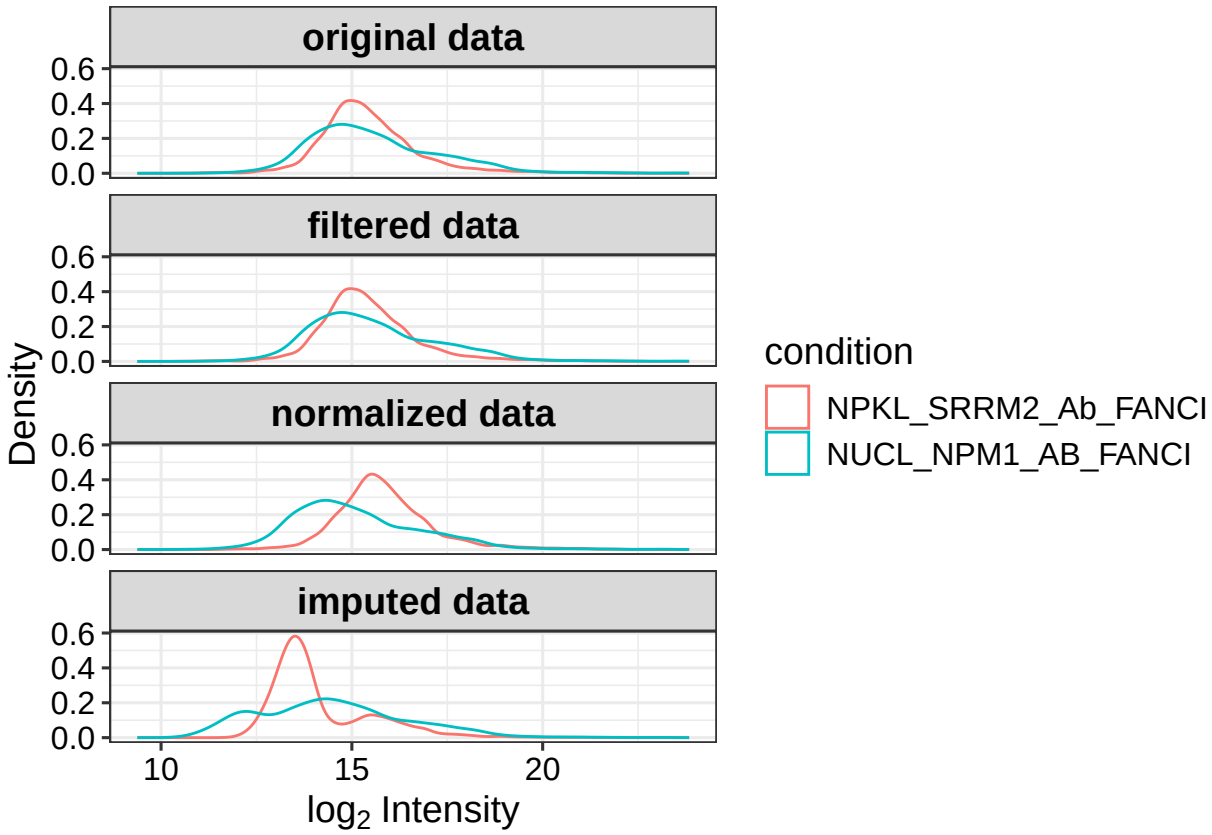
Missing value heatmap

A heatmap for proteins with missing value in each dataset. Each row represent a protein with missing value in one or more replicate. Each replicate is clustered based on presence of missing values in the sample.



Missing value distribution

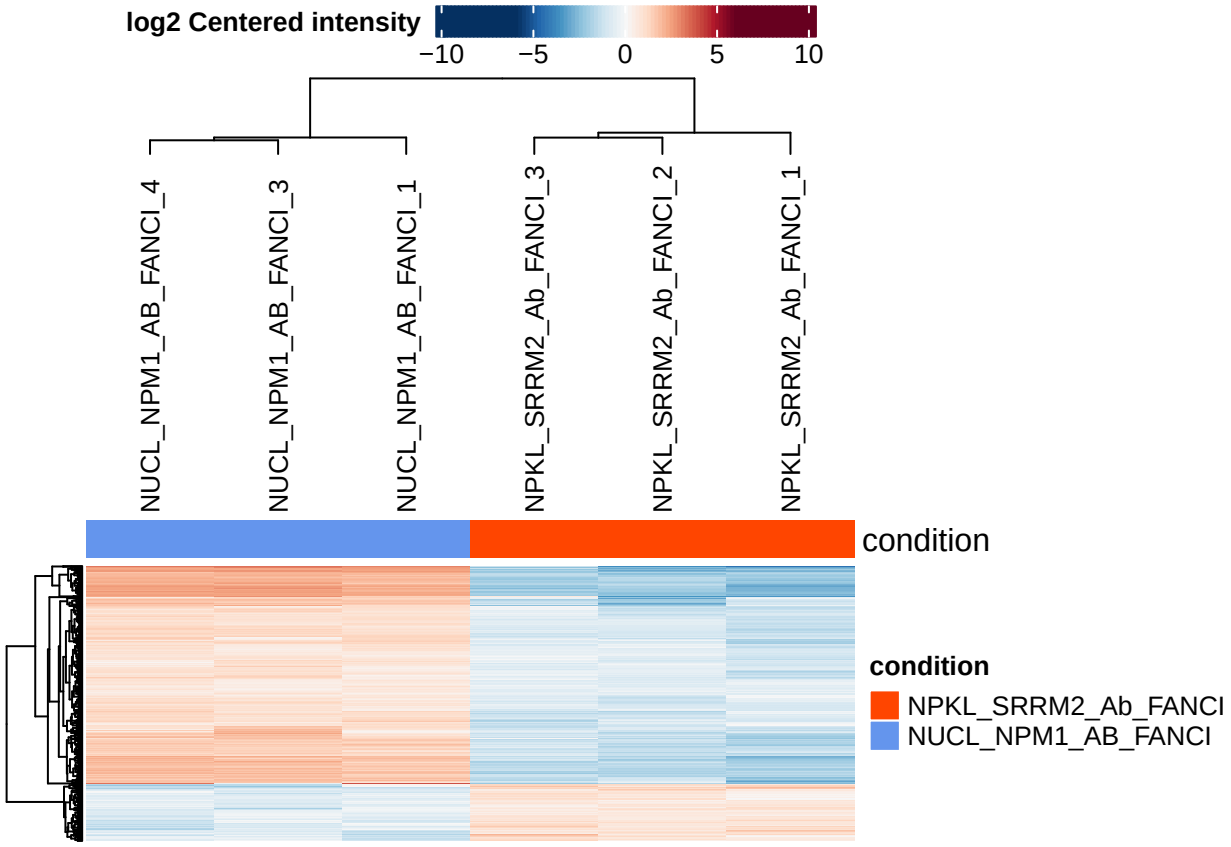
Protein expression distribution before and after imputation. The plot showing the effect of imputation on protein expression distribution.



Differential Expression Analysis (Results Plots)

Heatmap

A plot representing an overview of expression of all significant (differentially expressed) proteins (rows) in all samples (columns).



Volcano Plots

